

[illegible]

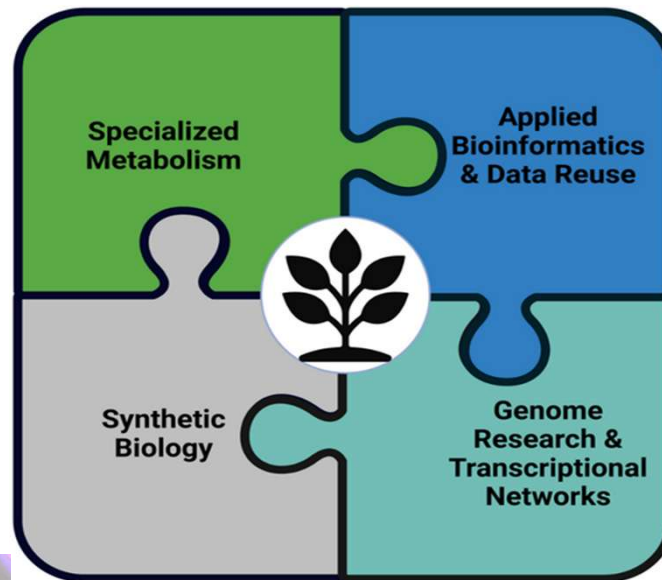
Samuel N. Meckoni & Nancy Choudhary

Command-line interface (CLI) basics for life scientists

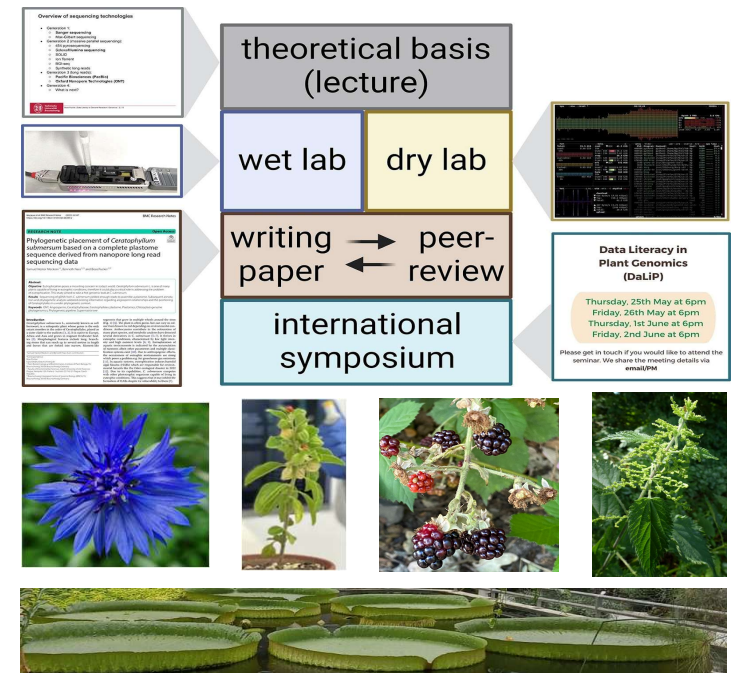
Plant Biotechnology & Bioinformatics (PBB)

Group overview

- Flavonoids:
Anthocyanins,
Flavonols, Flavones,
Proanthocyanidins
- Carotenoids
- Withanolides
- ...



Tools: KIPES, NAVIP, MGSE,
MYB annotator, DuplylCate...



Wolff *et al.*, 2023: 10.1515/jib-2023-0033 2

Dassow *et al.*, 2025: 10.1101/2025.01.31.635922

Course Overview

CLI basics for life scientists

- Basics of the Unix command line interface
- Unix file system
- Manipulating files and directories
- Combining bash commands
- Using these commands for biological insights



Samuel



Nancy



Boas

Availability of course material

- All materials will be shared at the end of this course.

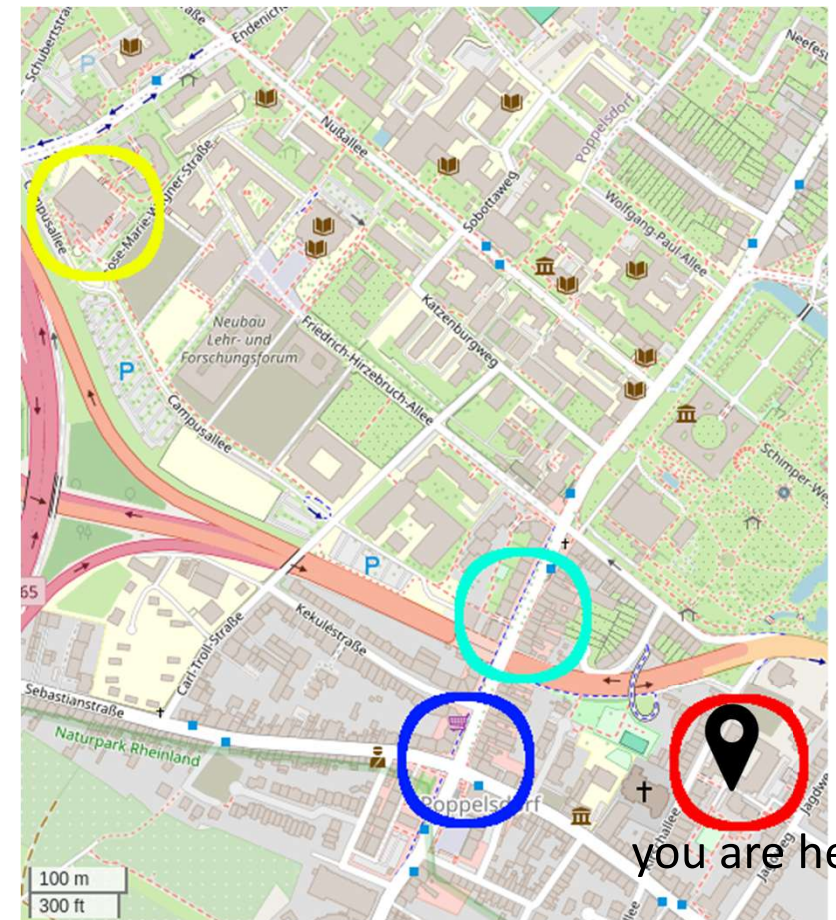
https://github.com/PuckerLab/teaching/tree/main/deNBI2026_CLIbasicsforlifescientists

Organisational I

- 10 AM -5 PM, 1 hour lunch break
- Tuesday: lunch break from 1 PM - 3 PM
 - PhD defense at 2 PM in 2.007
- Friday until 1 PM

Organisational II

- Mensa (15 mins walk)
- Local food stores
- Supermarkets



Agenda

- Introduction
- Theory
- Task 1.1
- Task 1.2

Introduction

- Study course/research field?
- What are your expectations?
- What is your previous knowledge?

Computerisation of the world

- specific task very efficiently
 - letter/word counting



Computerisation of the world

- specific task very efficiently
 - letter/word counting
- smart home/IoT
- useful for:
 - science
 - art
 - daily live
 - ...



Bioinformatics (Computational Biology)

- genetic sequences



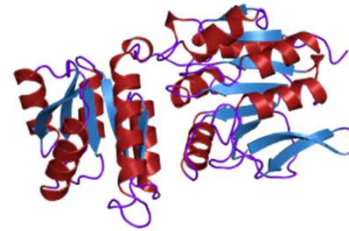
>> 100,000,000 bp
(genome sequence of a *Utricularia gibba*)



658,893 characters
(*'Little Brother'* by Cory Doctorow)

Bioinformatics (Computational Biology)

- genetic sequences
- structures



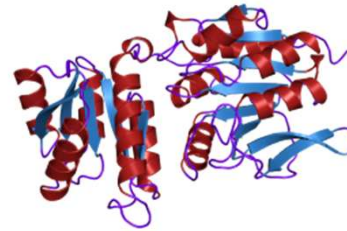
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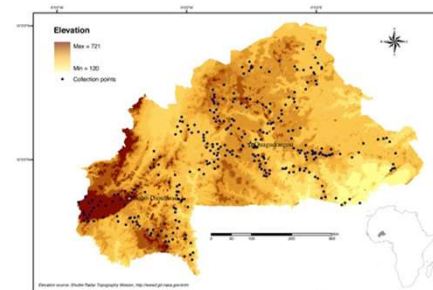
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Bioinformatics (Computational Biology)

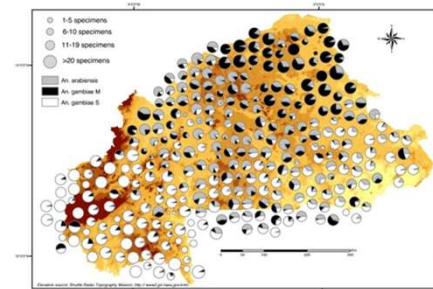
- genetic sequences
- structures
- networks and systems
- biodiversity
- ...



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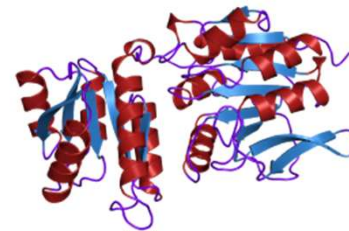


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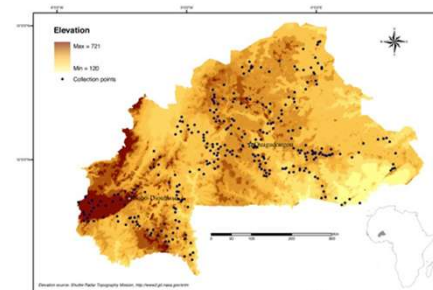


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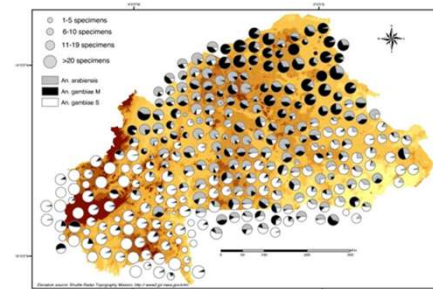
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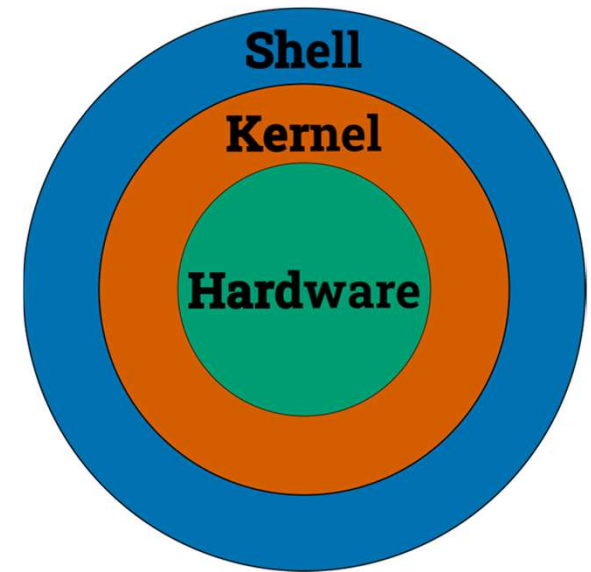


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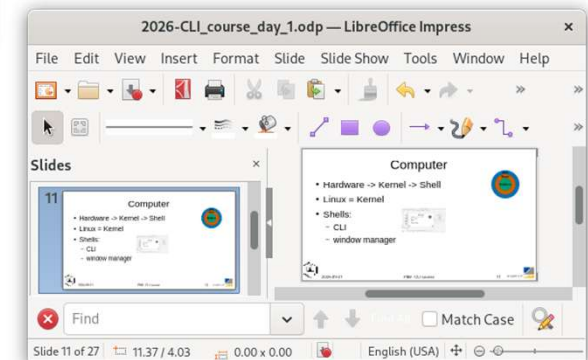
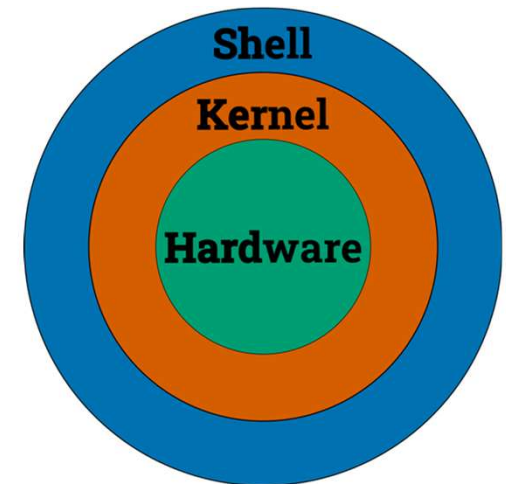
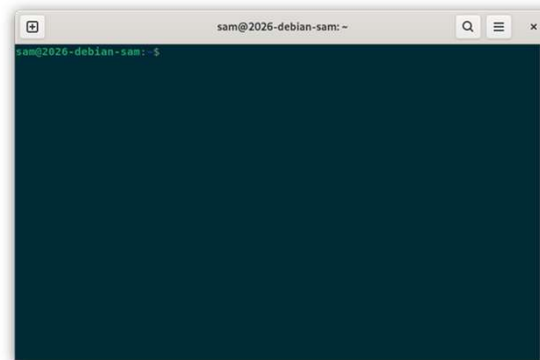
Computer

- Hardware -> Kernel -> Shell
- Linux = Kernel
- Shell

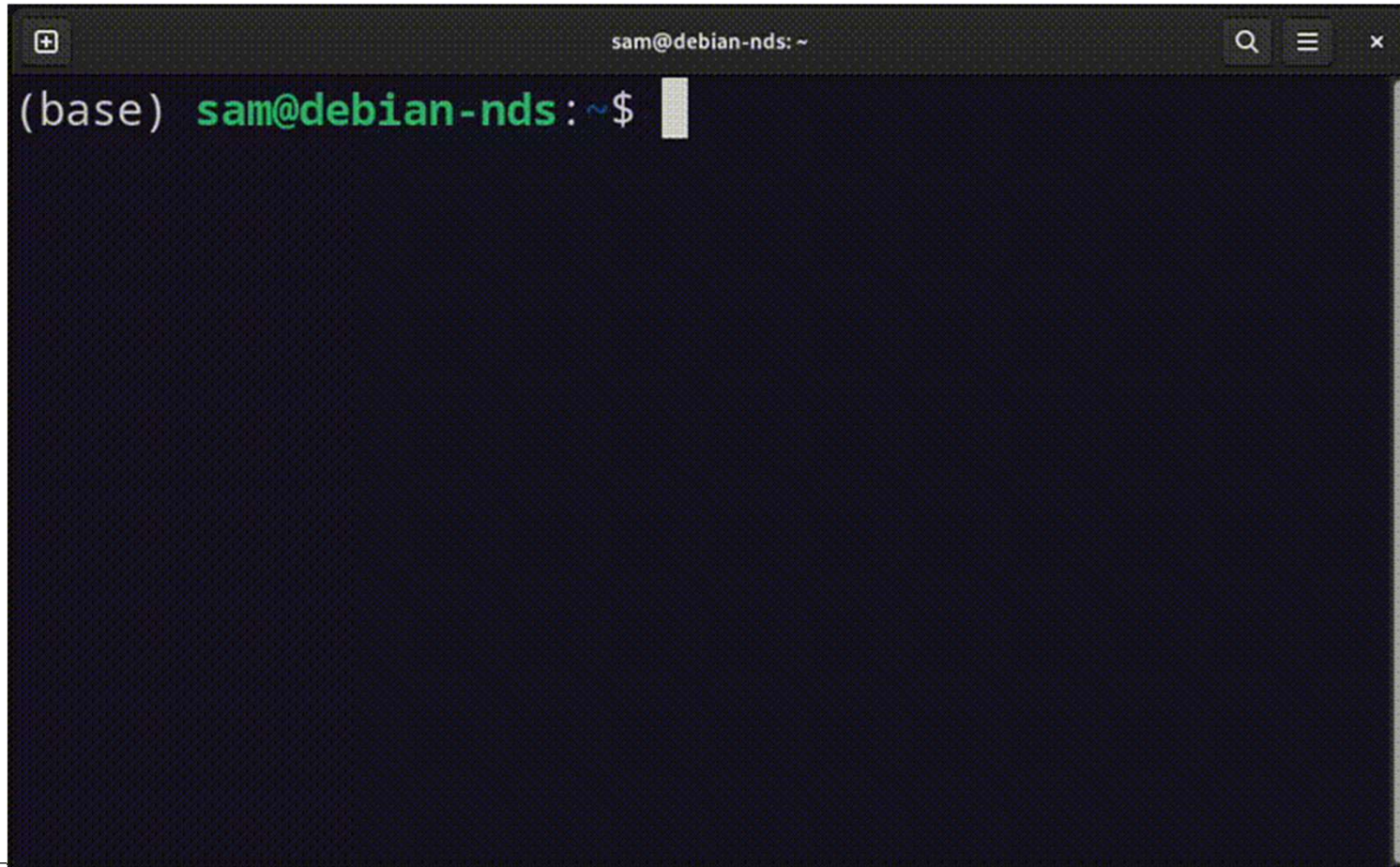


Computer

- Hardware -> Kernel -> Shell
- Linux = Kernel
- Shells:
 - CLI
 - window manager



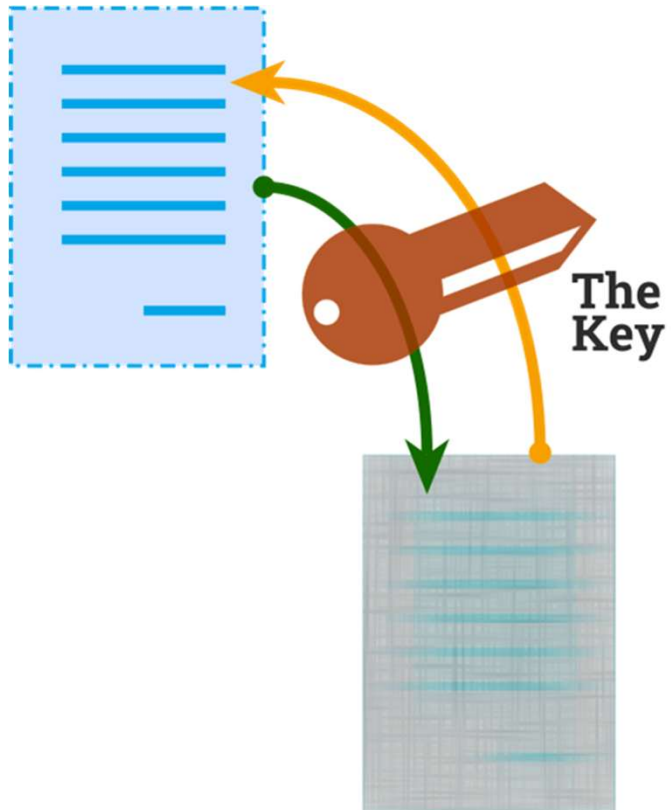
Terminal App



ssh = secure shell

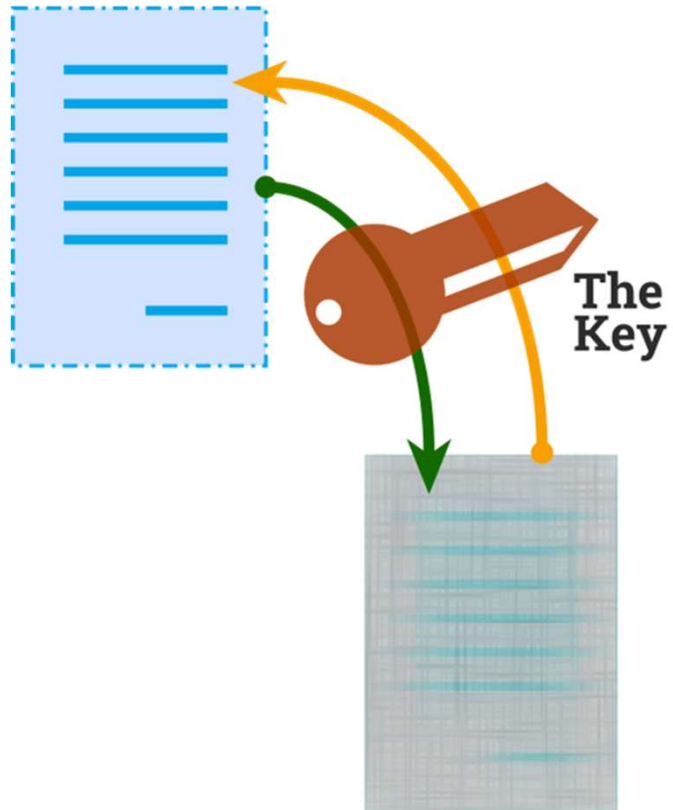
- cryptographic network protocol
- uses asymmetric encryption (“ssh keys”)

Encryption

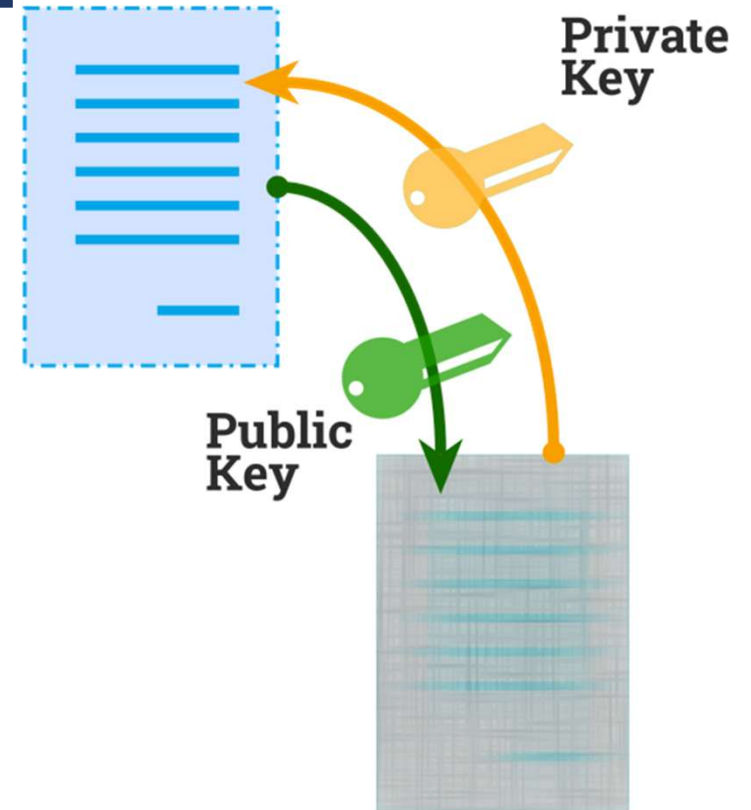


Symmetric encryption

Encryption

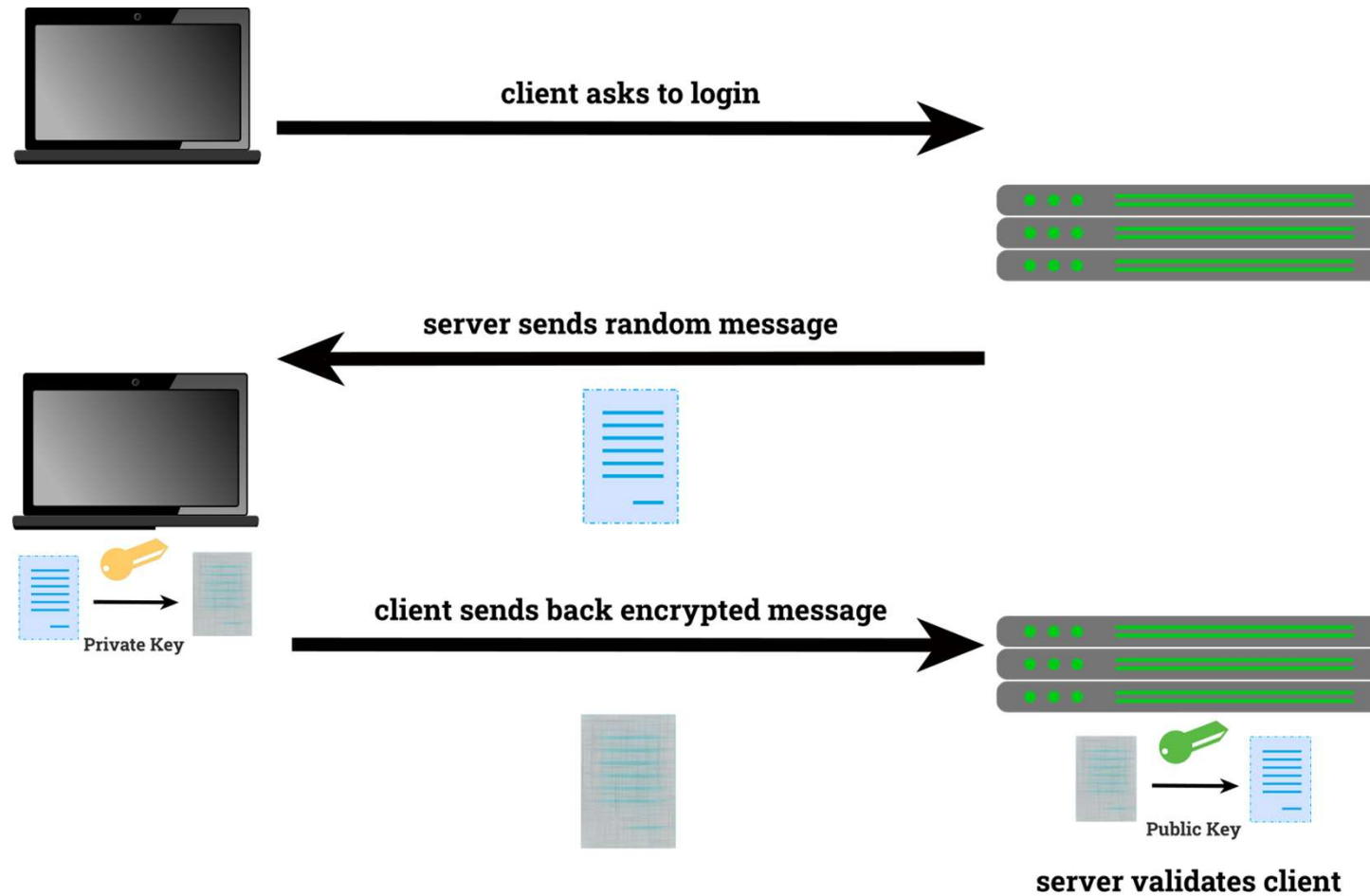


Symmetric encryption



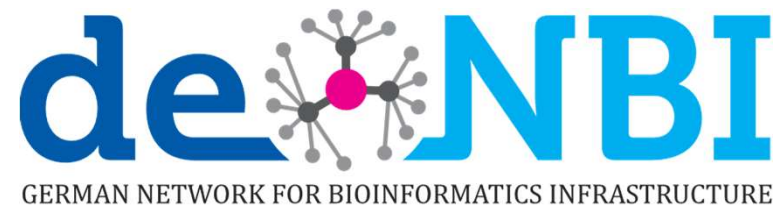
Asymmetric encryption

SSH

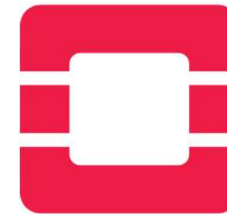
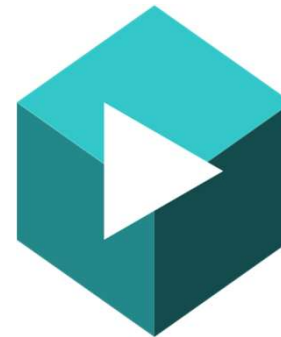


de.NBI

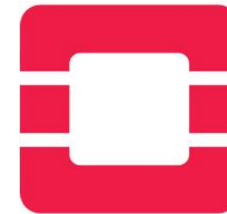
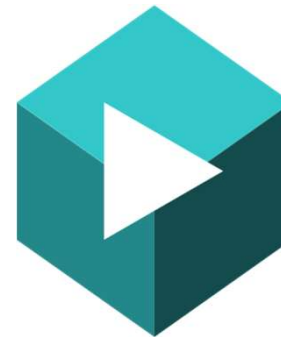
- German Network for Bioinformatics Infrastructure



- German Network for Bioinformatics Infrastructure
- Project types:
 - SimpleVM
 - Openstack
 - Kubernetes



- German Network for Bioinformatics Infrastructure
- Project types:
 - SimpleVM
 - Openstack
 - Kubernetes
- cloud.denbi.de
- simplevm.denbi.de



File paths I

- root folder: **/**
- home directory:
/home/ubuntu or **~**
(abbreviation for home directory)
- absolute file path:
/home/ubuntu/dir_2/abc.txt
- relative file path: **./dir_2/abx.txt**

current working directory, sets the relation for relative file paths



```
ubuntu@testsnm-7a4d2:~$ tree
.
├── dir_2
│   ├── abc.txt
│   └── def.md
├── directory1
│   └── file1
└── test.fasta
```

File paths II

```
ubuntu@testsnm-7a4d2:/mnt/test$ tree
.
├── dir_2
│   ├── abc.txt
│   └── def.md
└── directory1
    └── file1
```

File paths II

- absolute file path:
/mnt/test/dir_2/abc.txt
- relative file path:
./dir_2/abx.txt

```
ubuntu@testsnm-7a4d2:/mnt/test$ tree
.
├── dir_2
│   ├── abc.txt
│   └── def.md
└── directory1
    └── file1
```

Task 1.1: Login to denbi, create a VM and access it

- 1) Account registration: <https://cloud.denbi.de/wiki/registration/>
- 2) Joining the project: <https://signup.aai.lifescience-ri.eu/fed/registrar/?vo=denbi&group=202603pbbcli>
- 3) create ssh keypair: <https://simplevm.denbi.de> !
- 4) confirm unix name
- 5) create a vm:
de.NBI tiny, base image, 100 GB volume, if you like: a conda package
- 6) ssh command !

Task 1.2: Try out basic commands

- `cd`
- `ls`
- `man`
- `rm`
- `mv`
- `cp`
- `touch`
- `nano`

Do not use LLMs!